Chapter 3

Prosody, Scope, and Associativity

Abstract

In chapter 2, a cyclic algorithm is presented to derive the prosody of coordinate structures. In domains in which the associative law holds, a flat prosody is assigned; otherwise the prosody reflects the syntactic/semantic bracketing. This chapter presents evidence that the results obtained for coordinate structures carry over to other domains of the grammar.

3.1 Prosody in Coordinate Structures

Coordinate structures sometimes receive a 'flat' prosody, in which each conjunct is prosodically on a par, and sometimes a more articulated prosody, that reveals the syntactic bracketing as either left- or right-branching:

(1) a. $A \vee B \vee C$ — 'Flat' Prosody: A | ∨ B | ∨ C

b. $A \vee (B \wedge C)$ — 'Articulated' Prosody: A || ∨ B | ∧ C

c. $(A \vee B) \wedge C$ — 'Articulated' Prosody: A | ∨ B || ∧ C

In chapter 2, a cyclic mapping between syntax and phonology is presented to account for the prosody of coordinate structures.

The goal of this chapter is to show that the generalizations about prosody in coordinate structures carry over to other domains of the grammar. The following section summarizes the relevant properties of the theory of the syntax—phonology mapping proposed in chapter 2. Section 3.2 discusses associative domains other than coordinate structures, and shows they have a 'flat' prosody; section 3.3 shows evidence from domains that are not associative and in which scope crucially matters; and finally section 3.4 illustrates that some mismatches between prosody and syntax reported in the literature are only apparent and upon closer scrutiny give more evidence for the theory of syntax—prosody mapping developed in chapter 2.

3.2 Associative Domains

All and only structures with 'flat' prosodic structures are predicted to be associative domains, that is domains composed in a single cycle. The associative law holds when 2+ elements are combined in such a way that changing the bracketing would not change the meaning:

(2) Associative Law

$(\llbracket x \rrbracket)\llbracket y\ z \rrbracket = \llbracket x\ y \rrbracket\ (\llbracket z \rrbracket)$

The example discussed in chapter 2 are coordinate structures. This section presents additional examples of domains that are associative. Just as expected, they have a 'flat' prosody and are right-branching.

3.2.1 Predicate Sequences

Sequences of predicates in English appear to be right branching and yet they have a flat prosody. In the following example, the complement of each predicate is exactly everything that follows it:

(3) She wanted | to try | to begin | to plan | to move.

Each predicate can (but not need not be) accented. One factor in which predicate actually is pronounced with an accent is rhythm, as will be discussed in chapter 6. The prosodic grouping is similar to a coordination of predicates:

(4) to want, to try, to begin, to plan, and to move.

The predicate sequence receives a 'flat' prosody, i.e. the boundaries between the predicates are perceived as being equal in strength. Given the flat prosody, the expectation is that the domain is associative. But how can we interpret the predicates when the bracketing is changed?

A simple way to think about how predicates can be interpreted given a different bracketing is to think about the rebracketing in analogy to movement. E.g., the VP containing the lower two predicates can rightward move, λ-abstraction makes the the re-bracketed structure interpretable:

(5) ($\lambda x.$ wanted to try to begin x) ([$_{VP}$ to plan to move]).

This VP-movement leaves a remnant expression, and predicate abstractions turns it into a one-place predicate. The denotation of the final expression not changed by this restructuring, since the denotation of the moved VP semantically reconstructs to the complement position of 'begin' due to the λ-abstract. In other words, the law of association holds:

(6) $[\![A(BC)]\!] = [\![(AB)C]\!]$

In principle, this renders any predicate order interpretable:

(7) (λx. wanted x to begin to move) ($[_{VP}$ to try).

But of course this is not a grammatical word order for the sentence:

(8) *She wanted to begin to move to try.

The reason (8) is not grammatical is the movement step is ruled out by the constraints on rightward movement (Sabbagh, to appear).

That remnant sequences of predicates can be interpreted as one-place predicates can be motivated based on a number of grammatical phenomena. It is useful in the analysis of right-node raising. Consider first a simple case of coordination of predicates:[1]

(9) She wrote | and defended || her dissertation.

The prosody of this sentence suggests that the constituent 'her dissertation' attaches outside of the coordinate structure 'wrote and defended'. The boundary preceding the direct object is stronger than the prosodic boundaries within the conjunction, and is thus analogous to that of a left-branching coordinate structure:[2]

[1]It is not clear whether the boundary right before the right-node raised constituent has to be stronger than the one separating the conjuncts, but there is a clear intuition about a strong boundary.

[2]This is the case at least when the right-node raised constituent is accented. Hartmann (2001) observes cases were the right node constituent clearly is not preceded by a strong boundary. I assume that this is only the case when an unaccented constituent is right-node raised. This could be a pronoun or a head:

(1) a. John sáw | and Mary mét him

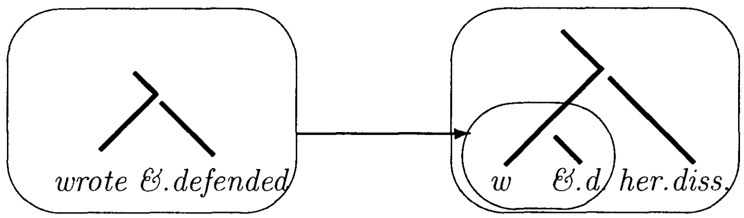
b. John had his glásses | and Mary had her lénses replaced

Hartmann (2001), Selkirk (2002), and Selkirk (to appearb), on the other hand, argue that even right-node raised constituents phrase with the material in the second conjunct. The predictions of this view with respect to durational effects as their are induced by boundary strength clearly differ then from the proposal here, and can be easily experimentally tested. This is an interesting question for future work.

(10) (A | and B) || or C

If the prosody in (9) is derived in the same way as that in (10), the coordinate structure 'wrote and defended' in (9) must constitute its own cycle before combining with its argument. The two functions are coordinated to create a conjoined function, which in the next cycle takes the direct object as its argument.

(11) Two Cycles



The first cycle is semantically interpretable: the remnant predicates form a complex predicate by λ-abstraction. The prosody furthermore suggests that in combining the coordinate structure with the direct object, the associative law does not hold, since the prosody clearly marks the structure as left-branching. Indeed, with a different bracketing, the meaning changes:

(12) She wrote || and defended | her dissertation.

This example involves a coordination of two VPs, one involving a predicate 'wrote' with an implicit argument and the other involving the transitive verb 'defend' and its object. Clearly, (9) and (12) differ in meaning.

Now consider the following example, an instance of right-node raising, analyzed as rightward movement in (Ross, 1967, Sabbagh, to appear):

(13) She wanted to begin | and then decided to postpone || her dissertation.

The prosody of this sentence suggests that the constituent 'her dissertation' attaches outside of the coordinate structure 'wanted to begin but then decided to postpone'. Again, the coordinate structure should constitute its own cycle. But this would imply that two apparent non-constituents are coordinated, namely 'wanted to begin' and 'then decided to postpone'.

In order for the coordinate structure to be semantically interpretable, the predicates have to be assigned a meaning, despite the fact that in each conjunct the direct object is missing. The idea is that they become one-place predicates by virtue of λ-abtraction:

(14) ((λx. wanted to begin x) and then (λx. decided to postpone x))

The entire structure in (14) constitutes a cycle and is combined with the object 'her dissertation' in a second cycle. Various different bracketings are in principle possible:

(15) a. She wanted | to try | to begin | to write.

b. She (wanted | to try | to begin) || to write.

c. She (wanted | to try) || (to begin | to write).

d. She (wanted) || (to try | (to begin | to write).

Just as in the coordinate structures discussed in chapter 2, the expectation is that they are only felicitous when the extra bracketing is motivated by information structure:

(16) She wanted to try to begin to write?

No, she wanted to try to begin || to procrastinate.

A sequence of predicates in which each predicate is the argument of the preceding one forms an associative domain, and is usually assigned a flat prosody. Flat prosodies are also observed in the domain of adjectives, which also can be viewed as sequences of functors:[3]

[3]But not all functor sequences are such that the emerging prosody is flat. Consider the difference between the following two structures:

(1) a. a (very | likely | false) || story. (a story that is very likely false)

b. a (very | likely) || false || story. (a story that is very likely but false)

When there is a left-branching node, the prosody of predicate sequences in (1) is more articulated.

(17) A dark | old | forest.

The combination of function application and λ-abstraction renders sequences of functors into associative domains. Just as in associative domains in coordinate structures, they receive a 'flat' prosody in which the elements are separated by prosodic boundaries of equal strength.[4]

3.2.2 Additional Arguments

Sentences can contain a variety of nominal arguments, receiving different thematic roles. Some of them are arguments of the main verb of the sentence, others are introduced by other heads; they are 'additional' arguments that are not part of the argument list of the main predicate. Consider the following sentence:

[4]An alternative view to the one involving variables and λ-abstraction is the one taken in categorial grammar. Steedman (1985, et. seq.) proposes that any sequence of predicates can be composed in more than one way. He uses the notion 'functional composition', an operation of meaning composition used in categorial grammar in addition to 'functional application'. In particular, Steedman uses a rule of forward function composition (Steedman (2001, 40), Steedman (2004)).

(1) Forward composition (simplified)
$X/Y : f \;\; Y/Z : g \Rightarrow X/Z : \lambda x.f(gx)$

Function composition, from now on abbreviated by the symbol '$\circ$', composes two functions f and g into a new function $f \circ g$. The definition makes sure that function composition does not change the meaning of an expression:

(2) $f \circ g(x) \equiv f(g(x))$

Function application assigns a meaning to a sequence of three elements A, B, and C if the meaning of C is in the semantic domain of B and B(C) is in the domain of A. Function composition allows the assignment a meaning to a rebracketed structure. This rebracketing does not alter the truth conditions of the outcome, i.e. , the associative law holds. Another domain in which forward composition is used in categorial grammar are predicate reorderings (cf. Steedman, 1985, 2000), to be discussed in more detail in chapter 6. The analysis of right-node raising in categorial grammar also involves forward composition (Steedman, 1985, et.seq.).

As observed in Sabbagh (to appear), the approach in terms of 'forward composition' in categorial grammar shares many properties with rightward movement accounts.

(18) Lysander baked a cake for Hermia.

Many theories treat the direct object as an argument of the verb, and the benefactive argument as being introduced by a different functional head. Part of the reason for making this difference is the intuition that a benefactive can be added to just about any sentence that involves an agent, while the thematic role of the direct object is closely tied to the meaning of a verb of creation such as 'bake'. Another reason for treating the benefactive as different is the fact that it is an optional argument.

The status of the subject in (18) is more controversial. Some theories of argument structure treat agents just like the direct object as an argument of the verb (e.g. Bierwisch (1983), Grimshaw (1990)); but there are also theories that treat agentive subjects as being introduced by a separate functional head, analogous to the benefactive.

One piece of evidence that subjects are not arguments of the main predicate is that the thematic role of the subject is not fixed by the verb, but depends on the combination of the verb and the direct object it combines with (Marantz, 1984). Schein (1993) presents a semantic argument for this view and argues that subjects are the argument of a separate event predicate. More arguments that agentive subjects are indeed introduced by a functional head other than the main predicate are discussed in Kratzer (1996) and Pylkkänen (2002).

The alternative to viewing an agentive argument as an argument of the main predicate is to assume an event predicate that takes it as an argument. This event predicate relates an event argument and a nominal argument, and was pursued within the 'Neo-Davidsonian' approach (Parsons, 1990) to argument structure. According to Kratzer (1996) and Pylkkänen (2002), agentive subjects are arguments of an inflectional *voice* head. Kratzer (forthcoming) furthermore presents evidence that an analogous analysis of the theme argument is not warranted, and direct objects should be treated as arguments of the verb.

I follow Larson (2005) in assuming that *voice* and any other Davidsonian predicates are tied together by existential closure, which binds the event variable in each of the predicates.[5] The event predicates are then combined by coordination.[6] The meaning of (18) is the following:

(19) $\exists e.[voice(e)(Lysander)\ \&\ bake(e)(cake)\ \&\ BEN(e)(Hermia)]$

A property of this analysis is that the combination of the event predicates obeys the associative law, just as in any other coordination or list structure:

(20) [[(voice(e)(Lysander) & bake(e)(cake)) & BEN(e)(Hermia)]] = [[voice(e)(Lysander) & (bake(e)(cake) & BEN(e)(Hermia))]]

The expectation is then that the elements of the list of additional arguments should be set off by prosodic boundaries that are on a par. This is compatible with speakers' intuitions about the prosody of these structures. Experimental tests, however, would be necessary in order to investigate this question further.

(21) Lysander | baked a cake | for Hermia.

If indeed the separate event predicates form an associative domain, the right branching conjecture predicts that they should form a right-branching structure. C-command evidence indeed shows that there is left-to-right c-command in sentences with several additional arguments:

(22) a. Every guest baked a cake for his host.

b. He reedited every movie for its main actor.

Pylkkänen (2002) identifies a number of arguments that are *not* introduced by a predicate that relate an event and an individual. Instead, they are introduced by a head that directly relates two nominal arguments. A case in point are the two

[5]The alternative would be to tie the predicates together by 'event identification' as in Kratzer (1996) or predicate modification Heim and Kratzer (1998), Pylkkänen (2002), which for the purposes here would not make a difference.

[6]I assume that the unpronounced list operator '*' is in fact involved here, discussed in chapter 2. Whether or not the coordinator is represented in syntax is not so clear: to be consistant with the discussion in chapter 2, I would actually have to claim that right-branching structures that lack functors are simply interpreted as lists conjoined with the set union operator '*'.

arguments in the double object construction in English, and low applicatives more generally.

Prosodically, they are not predicted to be on a par with the other arguments, but should form a domain of their own. I do not have the time and space to discuss the prosody of ditransitives, but refer the reader to Seidl (2000) and McGinnis (2002) for discussion of the prosody of different types of applicative constructions.

3.2.3 VP-Final Adverbs

VP-final adverbials seem to modify the meaning of entire VPs. Below is an example with 4 VP-final adverbials (Taglicht, 1984, 67):

(23) She saw him once, in 1939, outside the Albert Hall, after a concert.

One common analysis of VP-adverbials is that they take scope over the VP material preceding them, leading to a 'right-ascending' structure (cf. Andrews (1983), Ernst (2001)). Under the right-ascending view, VP-adverbials can be conceived of as functors taking VP-meanings as their arguments and returning an element of the same type.

If this right-ascending analysis were correct, then the structure would be left-branching. The prosody is then predicted to look as follows, at least if syntax-prosody mapping works analogous to coordinate structures. Left-branching structures necessarily show a fully articulated prosody:

(24) She saw him once, | in 1939, || outside the Albert Hall, ||| after a concert.

This does not match the intuitions that speakers have about the prosodic phrasing of this type of construction. The intuition about phrasing is that the prosody in the VP is 'flat', and the adverbs are separated by prosodic boundaries of equal rank.

How would the syntax have to look to obtain this prosody, if the theory proposed here is correct? It would have to be a right-branching, i.e. *right-descending* structure, and the domain should be associative, i.e. the bracketing should not matter in terms of the derived meaning. In other words, the structure should be parallel to an iterated coordinate structure.

(25) A, B, C, and D.

The parallel may not be too far-fetched. Taglicht (1984, 67) in fact *does* analyze sequences of VP-adverbials as coordinate structures. Similarly, Larson (2005) proposes to connect VP-adverbs with the VP by coordination, and argues that semantically, adverbs of this class are really event predicates. Just as in the case of additional arguments, they take an event variable rather than the entire VP as their argument. In fact, Davidson (1967) originally introduced event predicates precisely to account for modifiers. The denotation of a VP with several sentence adverbials looks as follows then:

(26) a. kissed Hermia in the forest for an hour.

b. $\exists$ e [kiss(L, H, e) & in-the-forest(e) & for-an-hour (e)].

Event predicates are added successively to the VP-structure. Larson (2005) takes them to be part of a coordinate structure in the scope of existential closure, which eventually ties them together to predicate over the same event. Existential closure is assumed to apply to the entire VP following Diesing (1992).[7]

Since the event predicates form an associative domain, they can combine within a single cycle. This correctly captures the fact that their prosody is flat. Based on the right-branching conjecture, the prediction is then that the structure of the VP should be right-branching.

Larson (2005) argues for a a right-descending analysis of VP structure. One piece of evidence for a right-descending analysis, as originally proposed in Larson

[7]An argument that the class of VP adverbs discussed here are not scope-taking comes from Phillips (2003), who observes in cases where three adverbs are stacked at the right periphery of the VP, the first two can take either scope.

(1) Sue kissed him many times intentionally in front of his boss.

Larson (2005) argues that the reason why the last predicate appears to take widest scope is due to the way the structure is split into a restrictor and a nuclear scope, and relates to focus and information structure. I will not discuss this problem here.

(1988), comes from NPI-licensing, which requires c-command. NPIs are licensed in VP-modifiers by the direct object and by preceding VP-modifiers:

(27) a. Lysander kissed nobody in any forest at any time.

b. John spoke rarely during any of our meetings. (Larson, 2005)

Variable binding also provides evidence for a right-descending structure:[8]

(28) a. Sue spoke to every child on his or her birthday.

b. John spoke during every session for as long as it lasted.

Under the theory of syntax-prosody mapping proposed here, the prosody of sentence-final adverbials is just as expected based on their syntactic structure, at least if the Larson's approach is correct.

3.2.4 Secondary Predicates

A second type of predicate can be analyzed analogously to VP-final adverbials: depictive predicates. Consider first subject-oriented secondary depictive predicates. They are prosodically similar to VP-Adverbs in that they do not induce strong prosodic boundaries in a sequence of predicates, and in that the order can vary.

(29) a. Hermia was dancing, completely drunk, without any fear, unaware of the abyss.

b. Hermia was dancing, unaware of the abyss, without any fear, completely drunk.

Not every order is equally unmarked. This is just as in coordinate structures and lists, where the order of conjuncts is preferred where it matches the order of the occurrence of events, of which event causes which, or other relations or scales that are used to connect the conjuncts in a meaningful way:

[8]More evidence from variable- and anaphor-binding is discussed in detail in Pesetsky (1995), Phillips (1996). Pesetsky (1995) also argues for a co-present second, right-ascending structure. I will not discuss the issue here, but refer the reader the discussion in Phillips (2003) and Larson (2005).

(30) a. open Monday, Tuesday and Friday.

b. big, bigger, biggest.

Pylkkänen (2002, 27) observes that depictive predicates also share semantic properties with VP-Adverbs. Similar to VP-adverbs, the state described by a depictive predicate must hold during the event described by the verb, i.e. they can be seen as predicates that attribute a property to an event. That secondary predicates indeed have to be eventive, Pylkkänen argues, is evidenced by the fact that individual-level predicates sound odd in depictive predications (as observed in Geuder (2000)):

(31) *? Hermia was dancing tall.

In Pylkkänen's analysis, depictives differ from VP-adverbs only in that in addition to the event argument, they also have an unsaturated individual argument of type e. The event argument, just as in the case of VP-adverbs, is bound by existential closure; the e-type individual variable is bound by a c-commanding nominal argument.

If secondary predicates form a list structure with the VP and form an associative domain, just as VP-adverbs do, then a right-branching right-descending structure is expected. Indeed, subject-oriented secondary predicates are c-commanded by the direct object, as is evidenced by variable binding, suggesting that they are indeed put together with the preceding VP by the default right-branching structure expected in a cycle:

(32) Mary Poppins rewarded every child convinced of his worthiness.

Under this analysis, the structure of the VP can be assumed to essentially that of coordination of predicates, which contain argument variables that are bound by c-commanding elements.

For secondary predicates, just as for VP-adverbs, the bracketing between multiple secondary brackets is immaterial—the associative law holds. Just as expected, a flat prosody emerges; and as expected based on the Right-Branching-Cycle conjecture, grammar imposes a right-branching structure on the domain.

Discussion

According to the argument in chapter 2 the domains that receive a 'flat' prosody in coordinate structures are precisely those in which the associative law holds. This section discussed four other cases of domains with a 'flat' prosody, and argued that the associative law also holds within these domains. The results from coordination carry over to other domains of the grammar.

In the present theory, the explanation for the 'flat' prosody between multiple constituents relates to the way they combine with the rest of the structure. If they are added in such a way that the associative law holds, they can be combined in one cycle, hence the resulting prosody is flat.

3.3 Prosody and Scope

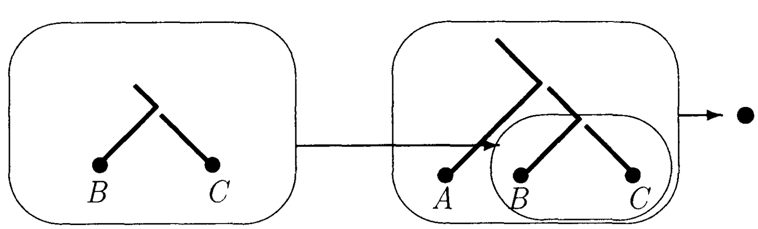
This section presents examples of ambiguities that parallel the pattern observed for coordinate structures in which the associative law does *not* hold. When elements are combined such that associativity does not hold, they take scope relative to each other. Depending on the bracketing, in the following structures 'and' has either wider scope than 'or' or narrower scope.

(33) a. A or (B and C)

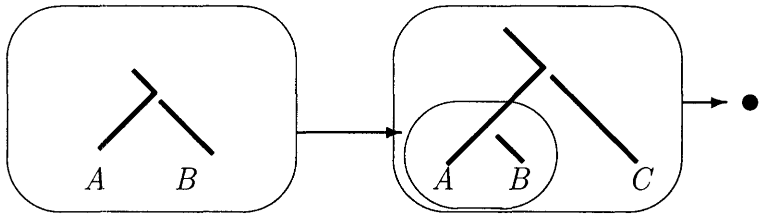
b. (A or B) and C

The break separating the constituent in the higher level of embedding is stronger than any boundary within the lower level of embedding.

(34) a. A || B | C:



b. b. A | B || C:



In other words, within each level of embedding, the conjuncts are separated from each other by boundaries of *equal rank*. If any conjunct is itself complex and contains further nested bracketings, the boundaries within that conjunct have to be of *lower ranks.* The syntactic grouping imposes a ranking of boundaries. The following generalization is derived:

(35) Scopally Determined Boundary Rank (**SBR**):
If Boundary Rank at a given level of embedding is n, the rank of the boundaries between constituents of the next higher level is n+1.

This has an effect on the relative scope of constituents: *An element within the lower level of embedding takes narrow scope relative to elements at the higher level of embedding.*

In the following I discuss evidence that the relation between prosody and scope observed in coordinate structure is also valid in other parts of the grammar.

3.3.1 Quantifier Scope

Taglicht (1984, 129) discusses an ambiguity in the scope of negative quantifiers.[9] I mark the material that negation takes scope over by underlining.

(36) a. I advised him to do nothing illegal, didn't I? *not* $<$ *advise*
I told him not to do anything illegal, didn't I?

b. I advised him to do nothing illegal, did I? *not* $>$ *advise*
I didn't tell to do something illegal, did I?

The tag-questions are just added in order to disambiguate the preceding ambiguous sentence. This section discusses evidence that prosody can (but does not have to) disambiguate these structures as well. The nature of the interaction between prosody and scope in those cases exactly parallels the case of simple coordinate structures.

Blaszczak and Gärtner (2005) observe that if a prosodic boundary intervenes between the quantifier and the higher predicate, the wide scope reading for the quantifier becomes impossible. The example Blaszczak and Gärtner (2005) give is the following:

(37) She has requested || that they read not a single linguistics book.
**not* $>$ *requested*; *not* $<$ *requested*

According to Blaszczak and Gärtner (2005) and Kayne (1998) wide scope is possible in embedded sentences, at least when subjunctive tense is used. The point can also be illustrated by an infinitival case:

[9] Taglicht (1984) also discusses similar facts for the scope of 'only'. According to Taglicht, the following ambiguity holds:

(1) a. They advised him to learn only Spanish. *only* $<$ *advise*

b. They advised him to learn only Spanish. *only* $>$ *advise*

I will only discuss negative quanitifiers here. The ambiguity of negative quantifiers is also discussed in Bolinger (1977, 48). Bolinger notes that taking wide scope is easier with negative quantifiers that he calls 'formulaic', such as 'no such thing', and 'no place at all'. These seem to be maximal elements on a scale, and possibly the fact that taking wide scope becomes easier is a reflex of the fact that negative quantifiers of this sort need to be focused and evoke alternatives.

(38) I will advise him || to do nothing illegal.

Another restriction on scope taking is that the quantifier has to be right-peripheral in order to take wide scope. Kayne (1998, 142) observes that if a negative quantifier is followed by a particle, it cannot take wide scope, but reports that it can when it follows the particle:

(39) a. I will force you to turn no one down.
$*not > force$; $not < force$

b. I will force you to turn down no one.
$not > force$; $not < force$

The analysis that Blaszczak and Gärtner (2005) proopose is that there is a restriction on scope-taking that makes reference to linear order and prosody:

(40) Condition on Extended Scope Taking (CEST)
Extending the scope of a negative quantifier $Q_{\neg}$ over a region σ requires σ to be linearly and prosodically continuous.

The relation between scope and prosody observed in the case of negative quantifiers is similar to that observed with scope ambiguities in coordination structures. A strong boundary following the first conjunct prevents wide scope of the 'or' in the following example:

(41) A || and B | or C $*or > and$; $or < and$

The analysis of the of case of negative quantifiers could be analogous to the one in coordinate structures if wide-scope taking involves a rebracketing:[10]

(42) (advised him to do) nothing illegal

[10]The scope ambiguities and their interaction with syntax and prosody discussed here suggest that scope-taking involves *overt* rebracketing. According to Kayne (1998), *all* scope-taking movement is overt, and covert movement does not exist. I will discuss a problem for this approach at the end of this section.

The prosody should then be parallel to a left-branching coordinate structure involving two cycles, with a boundary separating preceding the negative quantifier that is stronger than any boundary *within* the material it takes scope over.

(43) A | and B || or C

In order to derive the bracketing in (42), a first cycle would have to compose 'advised him to do', and spell it out. A second cycle composes the output of the first cycle with the negative quantifier.

Within the first cycle, the complement of 'to do' is not yet present, but the higher predicate 'advise' that in turn takes the infinitival phrase as its complement is. Blaszczak and Gärtner (2005) and Steedman (2001, 2004) use forward composition, as defined in (1), to analyze the data.

The alternative to use movement and λ-abstraction to explain how the meaning can be compositionally determined after rebracketing. Under either approach, the solution uses the same mechanism independently used for the various bracketings of predicate sequences discussed above.

In (38), 'advise him || to' cannot be a constituent to the exclusion of 'not a single linguistics book'—if it were, the prosodic break preceding the negative quantifier would have to be stronger than the one following the predicate it takes scope over, not weaker, as in the following structure:

(44) I will advise him | to do || nothing illegal.

The prosody in (38) is such that the 'advise him to do' cannot be a syntactic constituent. A similar rationale applies to the linear order restriction illustrated in (39). In order to take wide-scope, 'no one' should out-scope the underlined material, which would have to be composed in a first cycle. But this material does not form a syntactic constituent.

(45) I will force you to turn no one down.

A single cycle cannot comprise linearly discontinuous material, just as it cannot comprise a domain whose prosody reveals that it cannot form a well-formed level of embedding obeying the generalization in (35).

According to Blaszczak and Gärtner (2005), CEST is the restriction that prevents a wide-scope reading. In (38) it does so because a prosodic break intervenes, and in (39) it does so because the linear order condition is not met.

Even if the generalization in CEST holds descriptively, it may not be necessary to treat it as an active constraint in the grammar. An argument in favor of the view that CEST is not the driving factor comes from more complex examples. Blaszczak and Gärtner (2005) claim that a wide-scope reading in the following example is ruled out because there is a prosodic boundary following the relative clause. The CEST predicts that *any* intervening prosodic break should block a wide-scope reading.

(46) She requested that the students who finish first | read not a single linguistics book.

If the prosody reflects syntactic structure in the way proposed in chapter 2, the generalization should be a different one, namely that the break preceding the quantifier should be *stronger than any intervening prosodic break*. The reason is that all the intervening material must be in a deeper level of embedding in order for wide scope to be possible. This, according to the SBR, automatically entails that the prosodic break separating the quantifier from the VP must be stronger.

The following examples involve a relative clause intervening between the higher predicate and the quantifier, and yet wide scope is possible, at least if the prosodic boundaries in the underlined domain are not stronger than the boundary that sets off the negative quantifier. This goes against the prediction of Blaszczak and Gärtner (2005). An example involving a finite embedded clause is the following:[11]

[11] The example in (46) may be not constructed in a way that facilitates the wide scope reading. But the prediction is that if the prosodic breaks indicate the right bracketing, then the reading should be possible:

(1) She requested that the students who finish first read no additional linguistics book, did she?

(47) She <u>expected the students who failed the mid-term to take</u> none of the advanced tutorials, did she?

The tag-question should force the intended reading, and in fact it should also force a prosody that does not have a strong break after 'expected'. The same holds for the following example:

(48) She advised the new students that came in today to do nothing illegal, did she?

The prosodic fact that there can be no strong boundary between the predicate and the quantifier that takes scope over it can be related to the way syntax is mapped to prosody. The analysis presented here is similar to the approach to scope in categorial grammar Steedman (2004), and in particular to the proposal in Blaszczak and Gärtner (2005). It is different from Blaszczak and Gärtner (2005) in that it tries to derive the facts that CEST captures from syntactic bracketing and the way it maps to prosody instead of stating a constraint on interpretation that refers to prosody. Different predictions are made for examples like (47).

The discussion here unites the interaction between scope and prosody in the case of negative quantifiers with the facts observed on the coordinate fragment in chapter 2.[12]

The data discussed here suggests that scope *can* be taken by overt rightward movement (or alternatively by CCG-restructuring), which is directly reflected in prosody. This is not to say that scope *must* be taken by overt movement.

Fox and Nissenbaum (2000) give an argument that points to the possibility of covert QR in cases of negative quantifiers. They show that extraposing an adjunct from the direct object across a purpose clause forces a wide scope reading of the negative quantifier:

[12]Blaszczak and Gärtner (2005) also discuss the antisymmetric approach to the scope ambiguities in Kayne (1998), and propose a revision of this approach that is compatible with the data. I will not consider either proposal here, but refer the reader to Blaszczak and Gärtner (2005) for discussion.

(49) a. John must miss no assignment that is required by his math teacher in order to stay in school.

b. # John must miss no assignment in order to stay in school that is required by his math teacher.

c. John must hand in no assignment in order to stay in school that is required by his math teacher.

Sentence (c) shows that wide scope is possible under extraposition: it has a reading in which John is said not to have the obligation of handing in any assignment required by the math teacher. Sentence (b) shows that wide-scope is in fact obligatory. This sentence is infelicitous under a wide scope reading, but would be felicitous under a narrow scope reading as illustrated in (a); however, this reading is not available here.

The point is that 'no assignment' does not appear to attach higher than 'must' in this sentence, since it is followed by the purpose clause, which is presumably inside of the VP.

The analysis that Fox and Nissenbaum (2000) give this sentence involves covert QR of the negative quantifier and late merger of the adjunct. Covert QR might not be reflected in the prosody of a cycle. It would be interesting to investigate the prosody of these constructions and see how they square with the facts reported here.

3.3.2 Attachment Ambiguities

Taking scope interacts with prosodic boundary strength. Operators within lower cycles take narrow scope relative to operators within higher cycles. The discussion above showed that depending on whether or not disjunction takes wide scope over coordination, the boundary preceding 'or C' is either weaker or stronger than the boundary preceding 'and B':

(50) a. A | and B || or C — and < or

b. A || and B | or C — and > or

This section discusses data from the literature of on attachment ambiguities, and illustrates that again regularities already observed for coordinate structures apply.

Lehiste (1973) provided evidence that prosody disambiguates attachment ambiguities that relate to adjectives and their relative scope with respect to coordinated noun phrases.

(51) a. the (old men) (and women) — the old | men || and women

b. the old (men and women) — the old || men | and women

The results showed that the duration of words increased directly preceding stronger boundaries. Other correlates of strong boundaries found in Lehiste (1973) were laryngealization and insertion of pauses. A perceptual experiment showed that listeners are very good at telling the syntactic bracketing based on the prosodic cues involved.

The prosodic contrast in (51) is parallel to the contrast in the difference in relative boundary rank in the following two coordinate structures:

(52) a. A | and B || or C.

b. A || and B | or C.

An mirror image attachment ambiguity was studied in Cooper and Paccia-Cooper (1980, 34ff). Here, a modifier *followed* the constituent it modifies.[13]

(53) a. Lieutenant Baker instructed (the troop with a handicap).

b. Lieutenant Baker (instructed the troop) with a handicap.

The experimental measure in this experiment was the duration of the word preceding the boundary in question. More specifically, the duration of the 'key segments' in each example were measured, in the example above the duration of the segments /tru/ in 'troop' (i.e. the duration of the words excluding the codas). The difference between the structures was found significant.

The boundaries expected by the current approach are compatible with these findings, and are summarized below:

(54) a. Lieutenant Baker || instructed || the troop | with a handicap.

[13] I assume that 'with a handicap' is not an object-oriented depictive predicate as those in (3.2.4), but it a restrictive modifier of the NP 'troop' within the DP.

b. Lieutenant Baker || instructed | the troop || with a handicap.

The boundary following 'troop' in (a) is weaker than that preceding it; the reverse it true in structure (b).

More recent studies (Price et al., 1991, Wightman et al., 1992, Carlson et al., 2001) have shown further evidence that boundaries come in different strengths, and that stronger boundaries induce stronger lengthening effects than weaker boundaries. The relation between syntactic bracketing and prosodic boundary strength is similar to that observed in coordinate structures. The phonology and phonetics of attachment ambiguities in coordinate structures are further investigated in chapter 5.

3.4 Bracketing Mismatches

The prosody in coordinate structures showed a close match between prosody and syntax. Looking at other structures, a number of cases emerge where syntax and prosody seem to diverge in their bracketing.

It is important to note that there is no pre-theoretical notion of a mismatch in bracketing between syntax and prosody. Diagnosing a mismatch requires a theory of syntax that provides criteria to establish syntactic constituency; a theory of prosody that provides criteria for prosodic constituency; and a theory of the general mapping function from syntax to prosody, which provides an expectation how syntactic bracketing should map to prosodic bracketing. A diversion from this expectation constitutes a bracketing mismatch.

Under the theory of syntax—phonology mapping proposed in chapter 2 a structure in which the associative law holds receives a right-branching syntactic structure and a 'flat' prosodic structure:

(55) A, | B, | C | and D.

It seems, though, that the right-branching syntactic structure makes (C and D) a syntactic constituent, but the flat prosody does not make (C and D) a prosodic constituent. Does this not mean that there *is* a mismatch between syntax and prosody in (55)?

A mismatch would only exist if the prosody did not encode the syntactic bracketing, i.e. either encode the wrong constituent structure or not encode bracketing at all. But it is a general property of the mapping proposed here that right-branching structures map to flat prosodies (unless the associative law does not hold). Each domain with a 'flat' prosody' is mapped to a right-branching structure composed in one cycle, and vice versa. It is thus possible to retrieve the syntactic bracketing of (55), and it would hence be misleading to call this type of case a mismatch.

This section discusses what appear to be true mismatches between syntax and prosody, given the assumptions I made about syntax and prosody and the proposed mapping function. Upon closer inspection, the syntactic structure turns out to be just as expected based on the prosody, and what appear to be counter-examples to the theory proposed here turn out to lend it further support.

3.4.1 Coordinate Structures and Extraposition

Across various languages, a mismatch between syntax and prosody has been observed in cases in which a predicate takes a coordinate structure as its complement. The predicate tends to phrase with the first conjunct:

(56) (Predicate A) (and B)

An example is the case of Tiberian Hebrew, discussed in Dresher (1994, 19), who reports that fixed expressions, such as 'good and evil' in (57a), are phrased together. Otherwise, the predicate frequently phrases with the first conjunct (57b).

(57) a. (yōḏʕē) (ṭoḇ wārāʕ)
knowers (of)good and.evil (Gen. 3.5)

b. (kabbēḏ ˀet-ˀāḇīḵā) (wəˀeṯ-ˀimmeḵā)
Honor ACC-your.father and.ACC-your.mother (Deut. 5.16)

Phrasing in Tiberian Hebrew is reflected by phonological processes such as spirantization, which applies to post-vocalic stops within a phonological phrase. The underlined consonants are the ones that have undergone spirantization. But phrasing is also more directly noted in the Masoretic texts by an intricate system of accents, which according to Dresher reflect the prosodic structure of the verses.

The same phrasing seems to be possible in English. While it is possible to realize a stronger boundary after the verb, it is certainly not necessary, and (b) seems more natural:

(58) a. She kissed || Lysander | and Demetrius.

b. She kissed | Lysander || and Demetrius.

One context in which (a) is attested is the following:

(59) She hugged Hermia, and she kissed || Lysander | and Demetrius.

Phrasing a first conjunct with a preceding predicate looks like a genuine bracketing paradox. One possible solution discussed in Dresher (1994) is that the asymmetry could be related to XP-edge marking. The idea is that only left or right edges of XPs induce prosodic brackets, and that a language that has right-edge marking would predict the phrasing observed:

(60) (kissed Lysander)$_{\Phi}$ (and Demetrius)$_{\Phi}$.

Of course, this can only account for the phrasing in (57b, 58b), but not the phrasing in (57a, 58a).

Consider now the following cases, where again the predicate phrases with the first conjunct; the second conjunct follows the prosodic constituent consisting of the first conjunct and the predicate following it.

(61) a. She has (some blúeprints to leave) (and a book about Frank Lloyd Wright.)

b. (The students arrived), (and the professors as well.)

It is impossible in this case to phrase the second conjunct with the head:

(62) a. * (some blúeprints) (to leave and a book about Frank Lloyd Wright.)

b. * (The student) (arrived, and the professors as well.)

The phrasing follows straightforwardly from the approach in chapter 2. Extraposition rebrackets the string yielding a structure parallel to the coordinate structure in (52) and thus gives the correct boundary ranks.

The examples in (61) do not just involve a *prosodic* bracketing issue. It seems as if the predicate occurs *inside* of the coordinate structure. There is thus a purely syntactic bracketing issue: the verb ends up in the first conjunct, although it seems to take the entire conjunct as its argument. Any approach must analyze the construction as a case of syntactic extraposition (whether or not one resolves the bracketing problem by rightward movement or some other theoretical tool is a separate issue).

The bracketing in (61) is compatible with the edge-alignment approach, at least if [some blueprints to leave] is followed by an XP-break—which is plausible, since it could be the edge of the VP. But in order for this to work, the approach *also* has to invoke extraposition.

The major flaw of the edge-alignment approach is that it does not link the bracketing paradox to extraposition, at least for the case where the head precedes the conjuncts in (58). There is no reason to invoke extraposition here, in fact, edge-alignment is designed to derive a *mismatching* prosodic phrasing and assumes that the syntactic phrasing is unchanged. This, however, turns out to be false. A look at agreement illustrates why. I will first look at cases in which the predicate *follows* the first conjunct.

There is independent evidence that a syntactic rebracketing, and not just a prosodic rebracketing, is at stake. In structures like (61b), it is only the first conjunct that agrees with the following predicate:

(63) a. A student attends, and a professor.

b. * A student attend, and a professor.

If and only if the second conjunct is extraposed is first conjunct agreement possible:

(64) a. * A student and a professor attends.

b. A student and a professor attend.

A look at similar examples in German shows the same pattern. In the extraposed case, the verb agrees only with the first conjunct, while when the second conjunct does not extrapose, the verb agrees with both conjuncts:

(65) a. dass beim Seminar der Professor zugegen war/*waren, und die
that at.the seminar the professor there was/were and the
eingeschriebenen Studenten.
registered students

b. dass beim Seminar der Professor und die eingeschriebenen
that at.the seminar the professor and the registered
Studenten zugegen *war/waren.
students there was/were

This may be a sign that the second conjunct is really an adjunct when it is extraposed (cf. Munn (1993) for an analysis of coordination as adjunction). The grammar of extraposition/late merger of adjuncts is discussed in detail in Fox and Nissenbaum (1999) and Fox (2002).

The question is now whether there is any evidence for extraposition in the cases where the predicate *precedes* the coordinate structure, i.e. in the cases in (58)? Extraposition would be string-vacuous here. But there are cases where adverbials intervene which reveal that indeed extraposition is possible in English (Munn, 1993):

(66) John bought a book yesterday, and a newspaper.

There is also evidence for Extraposition from agreement facts. First, observe that verbs taking post-verbal subjects can agree with the first conjunct or with both conjuncts.[14]

(67) Yesterday at the seminar, there was/were a professor and two students.

Just as in German, extraposition forces first conjunct agreement:

[14]A complication is that some speakers use singular agreement at least in the present tense even with plurals:

(1) There's lots of them.

I am using the past tense, since the speakers I asked rejected singular agreement in the past tense in sentences with plurals.

(68) There was/*were a professor yesterday at the seminar and two students.

If first-conjunct agreement is really a test for extraposition, the prediction for English is now that there should be a correlation between the prosodic bracketing and first conjunct agreement, as summarized in the following paradigm, a prediction that is not easy to test just on impressionistic data:

(69) a. In the seminar room there were || a teacher | and two students.

b. In the seminar room there was | a teacher || and two students.

I cannot test whether the extraposition analysis of the apparent bracketing paradox presented in (58) applies for the Tiberian Hebrew example in (57). A restriction on phrasing in Tiberian Hebrew illustrates another parallel to the case of extraposition. Dresher (1994, 19) notes that when the predicate follows the coordinate structure, it is impossible for it to phrase just with the second conjunct:

(70) a. (kī-ḥemʾā ūd̲baš) (yōk̲ēl)
for-curds and.honey shall.eat

b. * (kī-ḥemʾā) (ūd̲baš yōk̲ēl)

The reason for this restriction is that there is no leftward extraposition of the first conjunct. Extraposition of the second conjunct on the other hand would lead to a structure in which the verb follows the first conjunct and phrases with it, as in the structure of (61). I do not know whether this latter kind of extraposition is available in Tiberian Hebrew, however.

The question of why extraposition applies and under what circumstances is a complicated one and cannot be discussed here in detail. Still, a straightforward prediction emerges from the discussion here: whenever extraposition is forced, the prosodic bracketing should follow accordingly, phrasing the predicate with the first conjunct; conversely, whenever extraposition is blocked, the coordinate structure should phrase together and be set off from the predicate by a stronger boundary.

Consider the following example due to Halliday, cited in (Clark, 1971). Here, the second conjunct must be a shorthand for an entire second sentence 'and I saw Mary, too'; otherwise combining it with 'too' would not make sense. The focus particle 'too' takes a propositional argument and presupposes that another proposition of a similar shape is true as well. So the second conjunct must stand for an entire proposition, rather than just an individual:

(71) a. I saw John yesterday, and Mary too.

b. ?? I saw John and Mary too yesterday.

This suggests that what I have called extraposition really involves a structure that is also semantically different from the unextraposed structure. The same is true for the case where the predicate follows the first conjunct. Only in the extraposed version can the focus adverb 'too' be used:

(72) a. Two people have arrived: John has arrived, and Mary, too.

b. * Two people have arrived: John and Mary have arrived, too.

As predicted based on the fact that 'too' forces extraposition, the *prosody* obligatorily signals the expected syntactic bracketing:

(73) a. I saw | John, || and Mary, too.

b. * I saw || John | and Mary, too

The case of extraposition in coordinate structure serves to illustrate that mismatches between syntax and prosody may sometimes only be apparent. The syntactic bracketing in this case may be just as expected based on the prosodic evidence.

3.4.2 Relative Clauses

One of the key observations that shaped theorizing on the prosody/syntax mapping is that clausal elements are mapped to their own intonational domain, independent of the syntactic bracketing. Consider the case of relative clauses (Chomsky (1965, 13), SPE, 372):

(74) This is | the cat || that chased | the rat || that stole | the cheese.

The relative clause forms its own phonological domain and is separated from the head of the relative clause by a boundary that is stronger than the boundaries that in turn separates the head from the predicate that precedes it. According to SPE, the bracketing in syntax should be as follows:

(75) This is [the cat that chased [the rat that stole the cheese.]]

The solution proposed in Lieberman (1967, 120) and in SPE was that a readjustment that rebrackets (75) into a different structure in which the three clauses are treated as on a par: "The resulting structure appears then as a conjunction of elementary sentences (that is, sentences without embeddings). This allows us to say that intonation breaks precede every occurrence of the category S (sentence) in the surface structure, and that otherwise the ordinary rules prevail" (SPE, 372).

Subsequent approaches took up this 'readjustment' account, but often considered the 'readjustment' to be an effect of the mapping of syntactic structure to prosodic structure, which result in a mismatch. The boundaries separating the relative clause can e.g. be taken to be due to alignment constraints that force prosodic boundaries at the edge of certain syntactic constituents, in this case a clausal node. The different bracketing is then seen as a genuine mismatch between prosody and syntax.

It is conceivable though that the readjustment takes place in *syntax*. The nature of readjustment was left open in SPE, and a 'performance' explanation was considered a possibility, although this was not fleshed out in detail. In this section, I will give some arguments in favor of a syntactic approach—the main point that I want to make is that there is evidence for a *syntactic* bracketing that *matches* the prosody.

In a natural rendition of (75) the boundaries after the verbs are usually weaker than those preceding the relative clause. This is exemplified in (76a), which is the preferred phrasing compared to (76b):

(76) a. that chased | the rat || that stole the cheese.

b. that chased || the rat | that stole the cheese

The prosody in (76b) may be appropriate in a context that has narrow focus on the direct object, but may otherwise not be the preferred phrasing:

(77) Who did the cat chase?

the cat chased || the rat | that stole the cheese

The phrasing that corresponds to the syntactic bracketing that Aspects and SPE assumed is then possible, at least under certain circumstances. But how to derive the apparently mismatching bracketing?

One possibility is that just as in the case of coordination discussed above, extraposition is involved. Extraposition of relative clauses is certainly possible:

(78) I saw the cat yesterday that caught the rat on Monday that had stolen the cheese on Sunday.

Extraposition would render a bracketing that corresponds to the prosodic bracketing positing in SPE:

(79) [[This is the cat] [[that caught the rat] [[that stole the cheese]]]]

In order to test whether prosody really mirrors the syntax, it would be necessary to control whether or not extraposition takes place or not. Adverbs can be used to force extraposition as in (78), but how can we prevent it from happening?

Hulsey and Sauerland (2005) argue that extraposition is impossible if the head of the relative clause is an idiomchunk:

(80) a. Mary praised the headway that John made.

b. * Mary praised the headway last year that John made.

If extraposition is involved in rendering the prosody in (75), then we expect to see an effect of idiom chunks on prosody. Consider the following two constructions, with the prosody that groups the head with the following relative clause:

(81) a. This was entirely due || to the advantage that he took || of the headway that she had made before.

b. This was entirely due || to the extra funds that he took || from the surplus that she had made before.

The prediction is now that the 'mismatching' prosody discussed in SPE should be impossible when the head of the relative clause is an idiom chunk. This is confirmed by impressionistic data collected from native speakers, who sense a contrast in acceptability between the following two examples:

(82) a. ?* This was entirely due to the advantage || that he took of the headway || that she had made before.

b. This was entirely due to the extra funds || that he took from the surplus || that she had made before.

This contrast constitutes evidence for the claim that readjustment, the rebracketing observed in SPE and Aspects, requires syntactic extraposition. There is no bracketing paradox between syntax and prosody in this case.

The judgment is reminiscent of cases with focus within idioms, which are infelicitous since there is no alternative for a focused material that would make sense as a replacement given the idiomatic interpretation of the structure:

(83) She *KICKED* the bucket.

It seems that one factor involved in a 'matching' (i.e. extraposition) derivation of relative clauses is that the head has to be focused. This would capture why idiom chunks are not allowed as heads of extraposed relative clauses.[15]

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[15]There is another property that might distinguish different types of relative clauses. Relative clauses derived by raising are often prosodically subordinated. This was observed i.a. in Bresnan (1971, 258):

(1) Mary liked the propósal that George left.

Nuclear stress falls on the head of the relative clause. But this, as was critically observed in Berman and Szamosi (1972) andn Lakoff (1972), is not always the case. Bresnan (1972, 337) illustrates the problem with the following two examples due to Stockwell (1972). When presented out of context, they differ in how they tend to be pronounced (of course, depending on context the stress could shift—How contexts influences prosody will be discussed in detail in chapter 2):

(2) a. Introduce me to the man you were tálking about.

b. I'll lend you that bóok I was talking about.

A comparison with German supports the extraposition-analysis of the English data. The OV word order in embedded clauses has the effect that extraposition is easy to diagnose—it is not string-vacuous as in English. Consider the following example, similar to the one discussed in SPE:

(84) Ich glaube dass dies die Katze ist, die die Ratte gejagt hat, die den
I believe that this the cat is that the rat chased, that the cheese
Käse gestohlen hat.
stolen has
'I think this the cat that chased the rat that stole the cheese.'

The relative clauses are extraposed, as is obvious considering the word order. The head is separated from its relative clause by the predicate. Without extraposition of the relative clauses, the structure would be a center-embedded structure, and end up unintelligible:[16]

(85) ?* Ich glaube dass dies die Katze die die Ratte die den Käse gestohlen hat gejagt hat ist.

According to Bresnan, there is also a semantic difference between the relative clauses: if the relative clause receives nuclear stress, then it is used to pick out one out of a set of alternatives defined by the head noun. In this case the head of the relative clause is a concealed partitive. In the sentence with nuclear stress on the head noun, however, there is no such partitive reading. I will present an potential explanation for the intuition reported by Bresnan in the discussion of effects of information structure in chapter 7.

The prosodic contrast between subordinated and non-subordinated relative clauses was related to raising *vs.* matching derivations of relative clauses in unpublished work by Karlos Arregi. Based on reconstruction effects, Arregi concludes that a raising analysis is applicable for relative clauses that are subordinated but not for those that contain the nuclear stress. If this analysis is correct, then extraposed relative clauses should not be prosodically subordinated, since extraposition forces a matching derivation. I have not tested this prediction.

[16]Center-embedding structures, used as an example for a performance restriction in Chomsky (1965), have been to shown to be hard to process in structures across languages and different constructions. The preference for extraposition of relative clauses my well have a motivation that relates to processing. This is in fact the explanation favored in SPE for the readjustment of relative clauses in (75). But this relation to processing factors does not mean that extraposition does not form part of syntax proper, and the syntactic and semantic effects of extraposition observed in Hulsey and Sauerland (2005) clearly show that extraposition is not just a phonological phenomenon.

Similar to the case of English, extraposition is impossible in German when the head of the relative clause is an idiom chunk.

(86) a. Peter war über den Bären den Maria ihm aufgebunden hatte
Peter was about the bear that Maria him given had
verärgert.
annoyed
'Peter was annoyed about the prank that Maria played on him.'

b. * Peter war über den Bären verärgert, den Maria ihm aufgebunden hatte.

The analysis of the extraposed relative proposed in Hulsey and Sauerland (2005) is that the relative clause is merged late as an adjunct, after the head of the relative clause has undergone covert movement. This is based on the analysis of extraposition of adjuncts in terms of 'later merger' discussed in Fox and Nissenbaum (1999) and Fox (2002).

The proposed solution to resolve the alleged mismatch between prosody and syntax in (75) is then exactly parallel as the solution for the case of the phrasing of the predicate with the first conjunct: syntactic extraposition is involved, and the prosodic bracketing matches the syntactic bracketing after all.

It may be that extraposed structures are sometimes preferred over non-extraposed structures for reasons that relate to parsability and maybe even pronounceablity, as discussed in SPE (p. 372) and also in Lieberman (1967, 120–121).

3.4.3 Possessor Constructions

One corollary of the way the cyclic system is set up is that all left-branching structures are prosodically fully articulated. This statement appears to be incorrect, however. Consider possessor-structures:

(87) John's brother's sister's dog's house.

This structure looks left-branching, but surely has a 'flat' prosody. The expected articulated prosody would look as follows, and is clearly very marked at best:

(88) John's brother's | sister's || dog's ||| house.

It is not clear though that the structure of possessors is underlyingly really left-branching.

(89) John's former house.

a. A former house that belongs to John (dispreferred).

b. A house that formerly belonged to John (preferred).

The NP in (89) has two readings, the more salient reading without further context being (b), the reading in which 'John' is in the scope of the adverb 'former', and the adverb modifies the NP 'John's house'. Larson and Cho (1999) argue that the left-branching structure gives wrong bracketing for this reading, and proposes that possessor constructions are underlyingly right-branching.

In Larson's analysis, possessor constructions have a structure comparable to the analysis Freeze (1992) gives to the verb 'have'. Freeze analyzes 'y have x' underlyingly as 'is x to y'. The locative preposition incorporates into the copula to create the complex auxiliary 'have'. Larson proposes that analogously, the possessor 's' is derived by incorporating a preposition into the determiner, so 's' is the equivalent of 'to.the':

(90) Movement analysis: the house to John $\rightarrow$ John to.the house

The underlying structure is compatible with building the complex possessor structure in a single cycle, since it is right-branching. The question of what triggers the surface reordering will be left open at this point. The facts from possessor constructions necessitate a looser notion of linearization of the constituents put together within a single cycle. For discussion of movement both preceding and following spell-out and a model of cyclic linearization see Fox and Pesetsky (2005), Ko (2005), and Sabbagh (to appear).[17]

[17]Another question that I will not address is the question of whether or not the associative law holds in these structures—which would be predicted based on their prosody.

Summary

This chapter discussed evidence that the findings from chapter 2 on the prosody of coordinate structures carry over to other domains of the grammar. Domains that are associative show a flat prosody, and in domains that are not associative the prosody reflects the syntactic bracketing that can be independently established looking at scope facts.

A look at apparent bracketing mismatches between prosody and syntax revealed further evidence for the mapping proposed in chapter 2. The prosody closely reflects the way structure is assembled during syntactic derivations.

This is not to say that there are *no* factors apart from syntax that shape prosody. Eurythmic factors e.g. can effect adjustments of the output grid structure. I will not explore these other factors here .